Unsupervised learning

Surprise me!

Finding the Unknown

- Find latent variables
  - Dimensionality Reduction / Representation learning
- Separate things
  - Clustering
- Find anomalies
  - Anomaly detection

A universal tool

- *Always* handy

- *Never* enough

Clustering

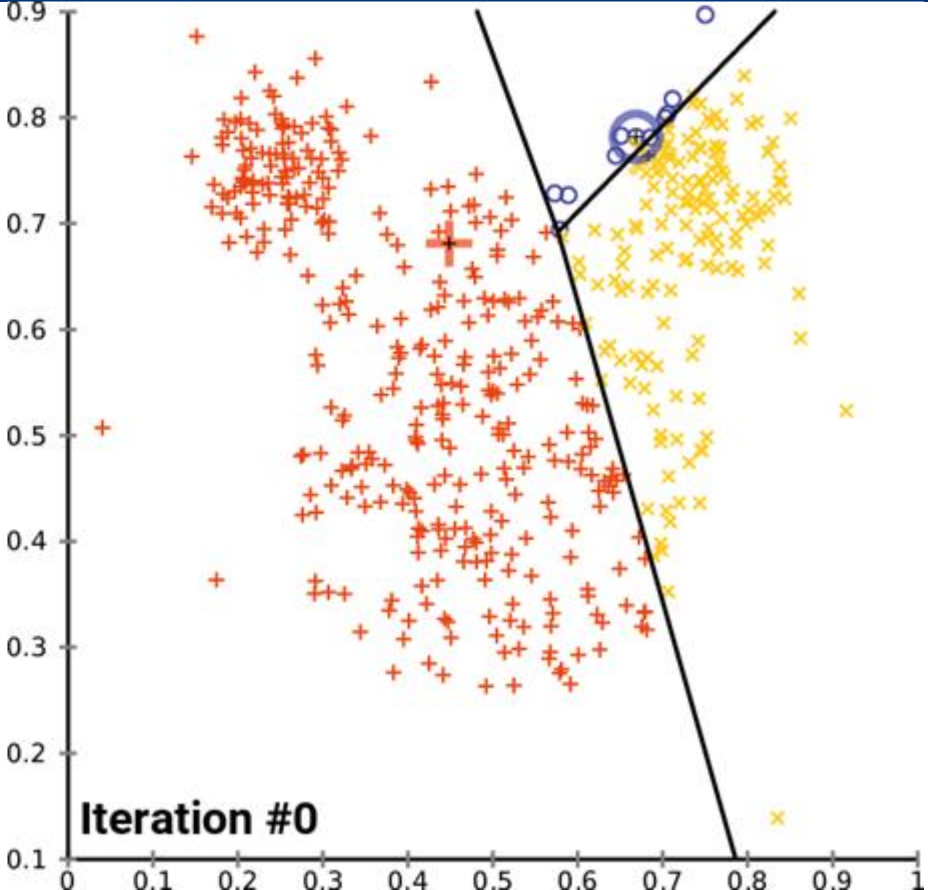
Labeling the unlabelled

K-means

Further details in ME/SM

1. Define random prototypes (centroids)
2. Attach all samples to the closest one
3. Recompute prototypes
4. Repeat 2.

K-means



The good and the bad

- No optimum guarantees
- Generally fast
- Sensitive to
  - number of clusters (k)
  - prototype initialization
- Spherical clusters (mean)

Cluster this

- (1,1), (1,2), (1,5), (3,2), (3,4), (4,1), (4,4), (5,3), (5,5), (6,2), (6,6), (7,7)
- Choose a random init
- Run 3 iterations of the algorithm

Variants

- k-Medians: sample in the middle
  - Actual data point. Outlier robust
- k-Medoids: sample *closest* overall
  - Actual data point. Outlier robust
  - Pair-wise similarity within cluster
- Improved initializations available (k-means++ spreads at init)

Hierarchical Clustering

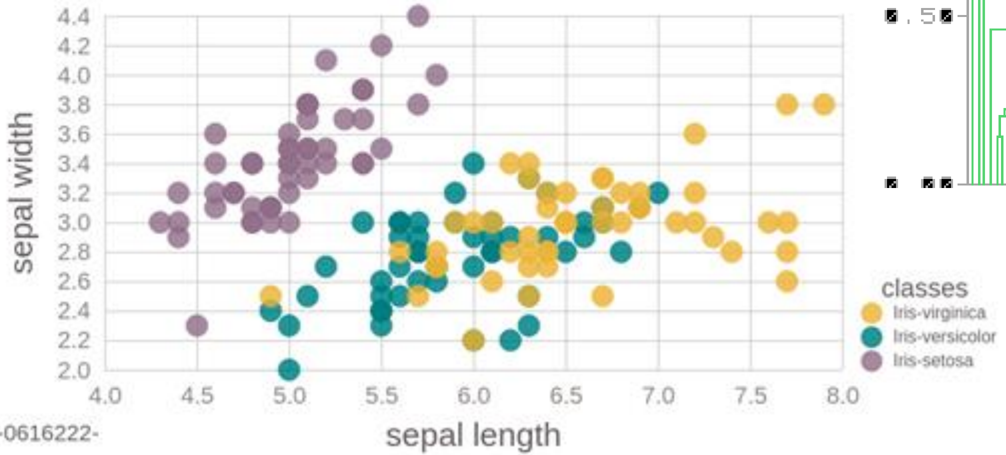
Further details in ME/SM

- Group things greedily, pair by pair
  - Bottom-up (agglomerative)
    - Start with N clusters
    - Merge two most similar from different clusters
    - Define new cluster
  - Top-down (divisive)
    - Split two most distant (centroids) from same clusters.
    - Define two new clusters. Assign others based on dist. to centroid
- Matrix of distances
  - Merging & updating

Dendograms

Further details in ME/SM

- Y axis = distance
- Branch length
  - Line cut



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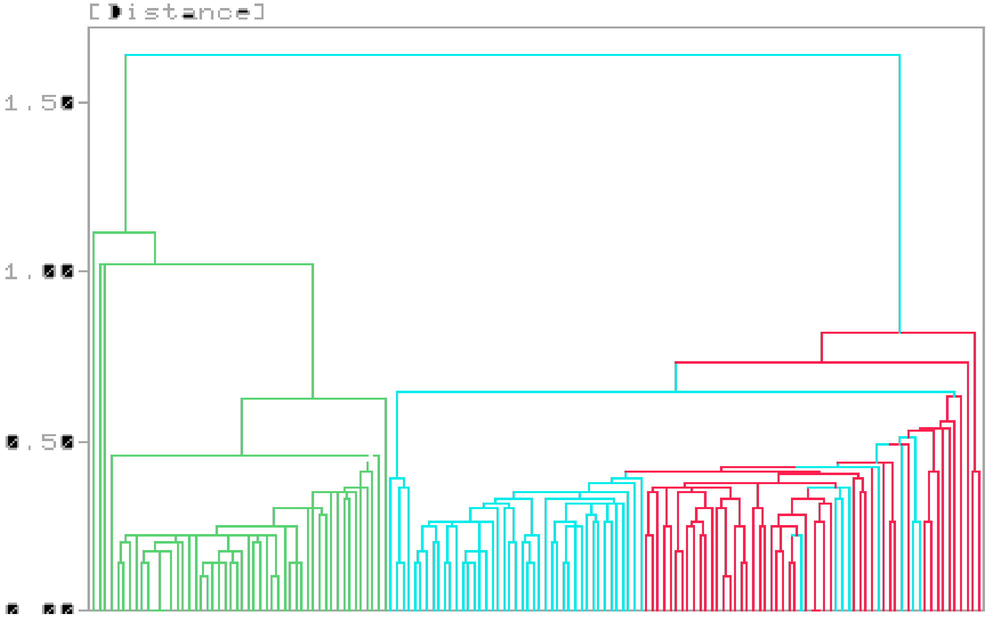
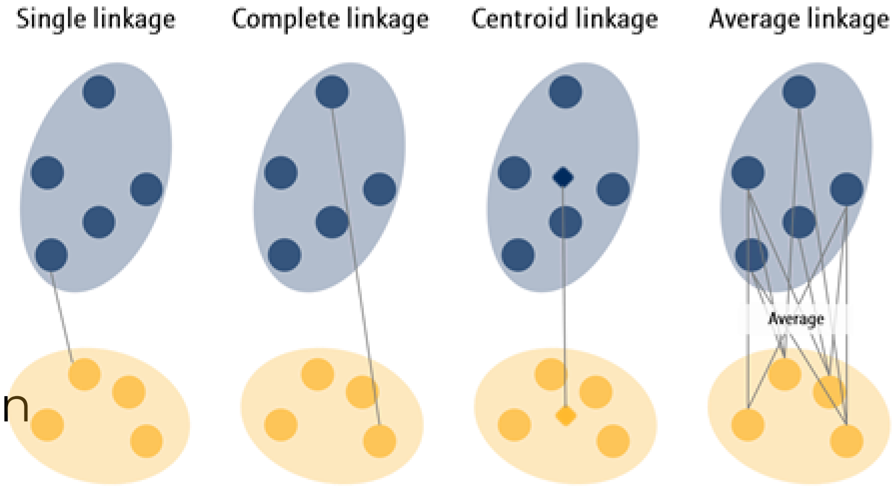


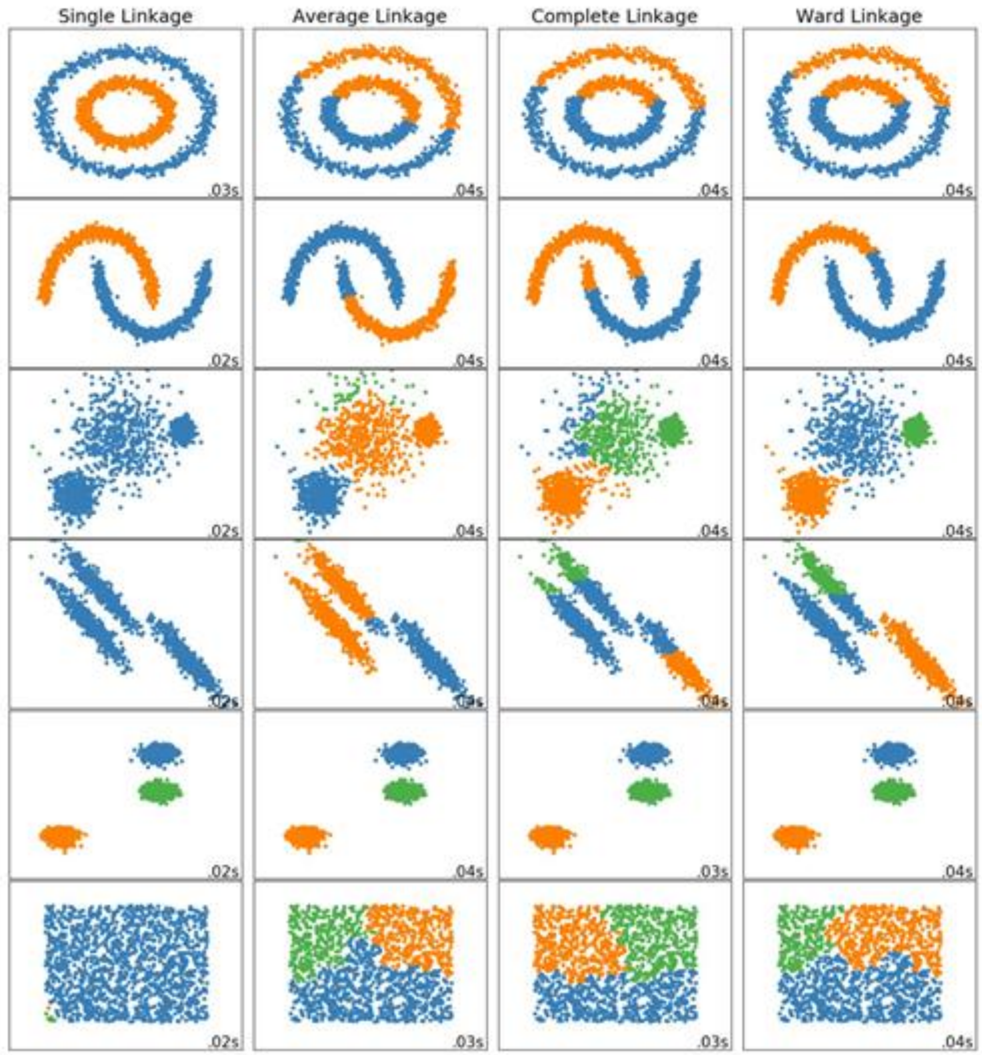
Hierarchical Clustering

- Linkage type (distance between clusters)
  - Complete (max, spheres)
  - Average (mean, spheres)
  - Single (min, ladders)
  - Ward (minimum variance within cluster)



Hierarchical Clusterir

- Linkage type
  - Complete (max, spheres)
  - Average (mean, spheres)
  - Single (min, ladders)
  - Ward (variance)



The good and the bad

- Expensive (recompute dist. mat.)
- Sensitive to linkage type
- Sensitive to outliers & noise (single linkage!)
- Usable with any similarity measure
- Interpretable
- Kinda k-free

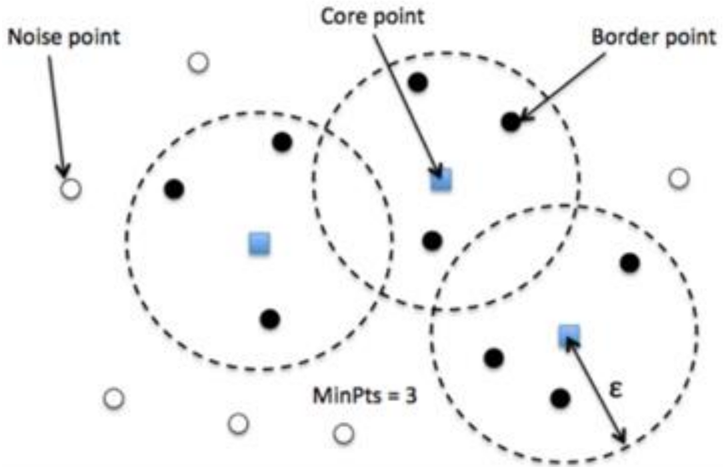
Cluster this

- (1,1), (1,2), (1,5), (3,2), (3,4), (4,1), (4,4), (5,3), (5,5), (6,2), (6,6), (7,7)
- Choose a tie breaker
- Build the dendogram

DBSCAN

Further details in ME

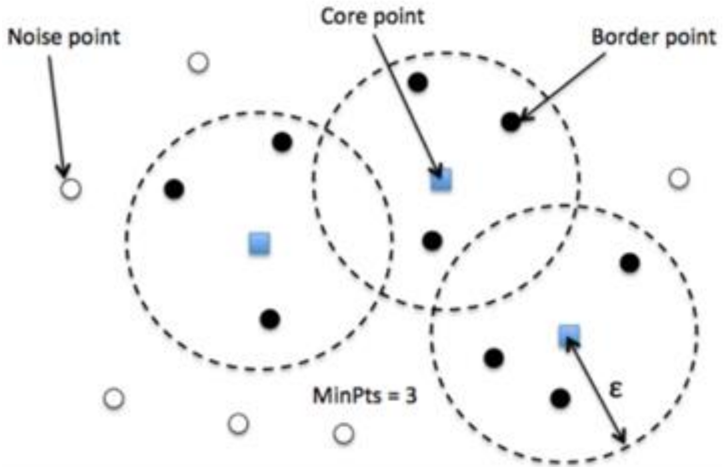
- Find dense sample regions
  - Core point: Potential cluster center (min. neighb. at max. radius)
  - Border point: Neighbour to a core point
  - Noise points: Others



- Label all nodes
- For each core point
  - start or expand a cluster
  - propagate label to border points

DBSCAN

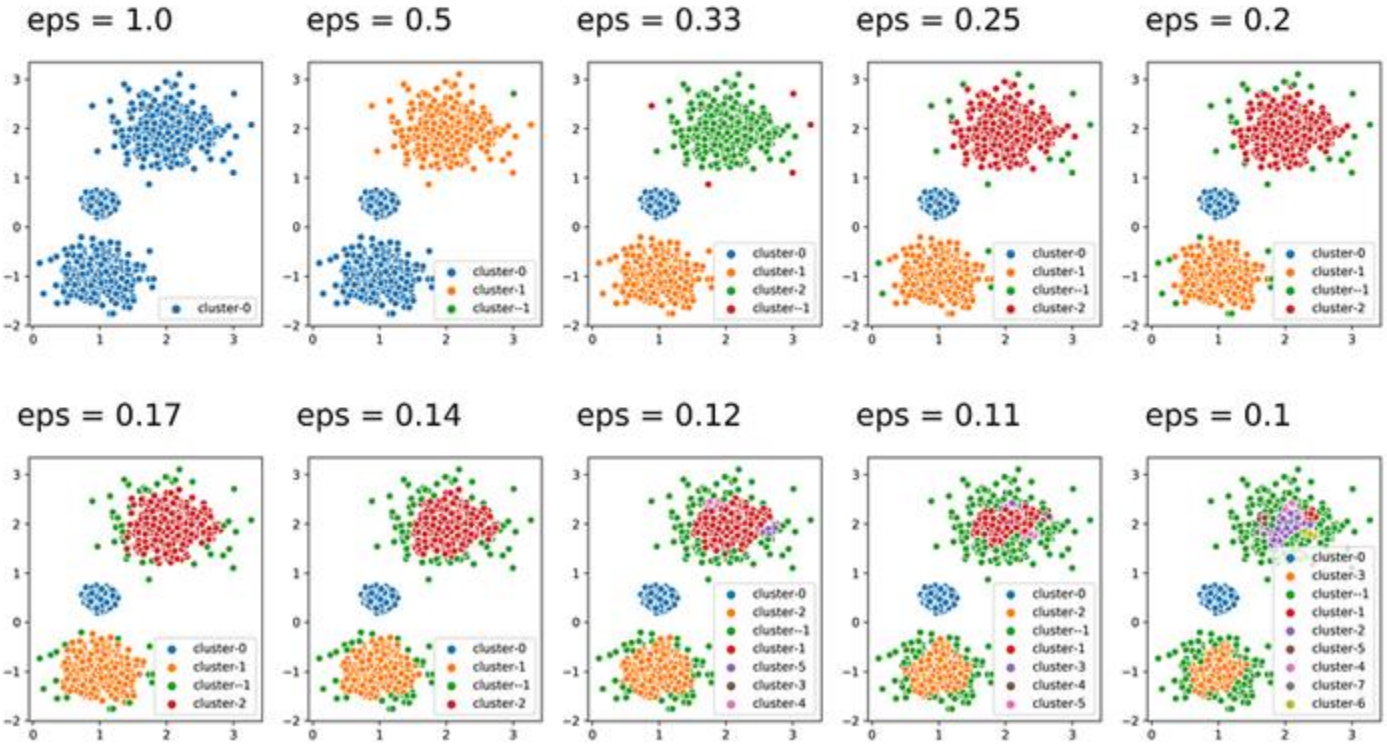
- Density-based clusters
- Noise: Unassigned samples
- Very sensitive to hyperparams
- Finds one k



DBSCAN params

- Radius (epsilon)
  - High: Larger clusters. Irregular.
  - Small: Smaller clusters, more noise points. Spherical
- Num. min. points
  - High: Denser clusters
  - Smaller: Sparser clusters

DBSCAN params



The good and the bad

- Bad for variable density datasets
- Works with any pair-wise similarity
- Fast
- No k needed (radius and min. points instead)
- Robust to outliers
- Non-deterministic for ties or border samples
- OPTICS, HDBSCAN (Hierarchical, using the radius)

Anomaly Detection

It's weird

Combination of other stuff

- Statistical measures
- Density metrics (KNN, DBSCAN)
- Unsatisfied frequent itemsets
- …
- Infrequent due to data volume